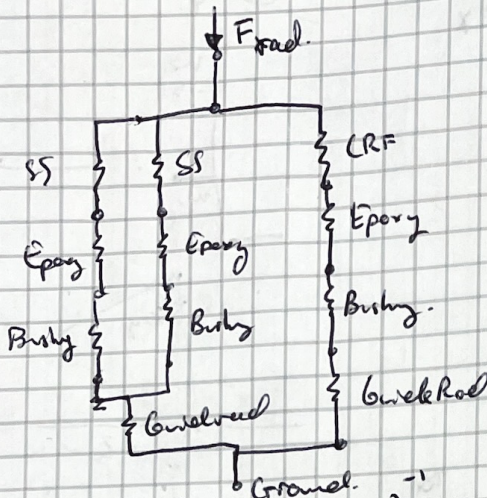


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$$k_{LHS} = \left(\frac{2}{\frac{1}{k_{SS}} + \frac{1}{k_{E_{poxy}}} + \frac{1}{k_{Bushing}}} + k_{Gusset\ rod} \right)^{-1}$$

$$k_{RHS} = \left(\frac{1}{k_{CRF}} + \frac{1}{k_{E_{poxy}}} + \frac{1}{k_{Bushing}} + \frac{1}{k_{Gusset\ rod}} \right)^{-1}$$

To simplify assume $k_{Bushing} \& k_{E_{poxy}} \gg k_{SS} \& k_{CRF}$

$$\Rightarrow k_{LHS} = \left(\frac{2}{2k_{SS}} + \frac{1}{k_{Gusset\ rod}} \right)^{-1}$$

$$k_{RHS} = \frac{k_{CRF} k_{Gusset\ rod}}{k_{CRF} + k_{Gusset\ rod}}$$

$$k_{Gusset\ rod} = \frac{192EI}{L^3}, \text{ for steel}$$

& k_{SS} = column of steel above the rail stiffness $\approx$ column of Aluminum.

$$k_{CRF} = \frac{\partial F_x}{\partial x}$$

For simplicity assume $k_{CRF} = AE/L$ which gives: $\frac{\sigma_{max} \times \omega_{max} \times E_{max}}{L_{max}}$

This then gives us the following

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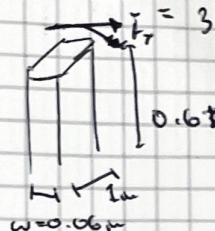
All Dimensions taken from assembly CAD.

Given the same values for these machine elements we get

$$k_{\text{Guide rod}} = \frac{192 \times 200 \times 10^3 \times 253.24 \times 10^{-3} \times 6.478 \times 10^{-9}}{(223.24 \times 10^{-3})^3} \quad \leftarrow \text{Length of rod from CAD}$$

$$= 22.36 \times 10^6 \text{ N/m.}$$

$k_{\text{Flexure}} =$ Assume this is a simply supported cantilever beam



$$= \frac{3EI}{L^3} = \frac{3 \times 68 \times 10^9 \times 60.9 \times 10^9 \times (0.06^3 \times 1/12)}{(0.64 \times 25.4 \times 10^{-3})^3}$$

$$= \frac{3 \times 3072 \times 10^{12} \times (7.5 \times 10^{-12})}{4.3 \times 10^{-15}} \times 297.34.$$

$$= 1.1 \times 10^6 \text{ N/m.}$$

$k_{ss} = 3EI/L^3 = \frac{3EI \times \text{width} \times \text{dist}^3}{12 \text{ bolt}^3} = 1.6 \text{ k N/m/m}$
(assume rectangular area above Guide rod nuts)

⇒ Both steel experience the same deflection but different forces

$$k_{\text{RHS}} = \frac{1.1 \times 10^6 \times 22.36 \times 10^6}{(1.1 + 22.36) \times 10^6} = 1.05 \times 10^6$$

$$k_{\text{LHS}} = ((2 \times 1.6 \times 10^6)^{-1} + 1/22.36 \times 10^6)^{-1} = 2.8 \times 10^6$$

$$\Rightarrow F_x = F_{\text{rod}} \times \frac{k_{\text{RHS}}}{k_{\text{LHS}} + k_{\text{RHS}}} = F_{\text{rod}} \times \frac{1.05}{2.8 + 1.05} = 0.27.$$

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